Tungsten in Polymetallic Nodules

Global occurrence, geochemical enrichment and priority exploration regions

A technical screening report on marine ferromanganese nodules and cobalt-rich crusts, with a ranked assessment of the strongest global targets for Μετάλλιον Inc.

59 ppm	149 ppm	287 ppm
Cook Islands nodule mean W (26-sample legacy dataset; [18])	Highest nodule assay in the cited (limited) Indian Ocean sample set	Maximum W reported in Emperor Chain Fe-Mn crusts, not nodules

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Prepared for: Μετάλλιον Inc.

IMPORTANT DISTINCTION

Tungsten is documented in oceanic Fe-Mn nodules and crusts, but usually at trace-to-minor concentrations. The report therefore distinguishes the best nodule province from the highest tungsten assays in other marine Fe-Mn deposits.

Executive assessment

Principal conclusion

Recommendation

Prioritise the Cook Islands EEZ as the best nodule-first exploration jurisdiction and regional geological target. Focus on the Penrhyn Basin and the northern portion of the mapped north-south high-abundance corridor. Treat tungsten as a potential by-product of a broader Co-Ni-Cu-Mn-Ti-REE system, not as the primary ore. In parallel, run a tightly scoped Indian Ocean verification study around the published high-W nodule sample CP8130.

The highest tungsten concentrations identified in the literature do not point to one simple “tungsten nodule province”. The best published nodule assays are in a limited Indian Ocean dataset (up to 148.5 ppm W). Cook Islands nodules have a reported mean of about 59 ppm W in a 26-sample dataset, but the province combines high nodule abundance, high cobalt and a published estimate of 30,215 tonnes of contained W in one selected high-abundance area. The Emperor Seamount chain reaches up to 287 ppm W, but those are Fe-Mn crusts coating guyots, not free nodules. Recent S-K 1300 disclosures for Cook Islands EL1 and EL3 report project-level nodule resources, but their resource tables do not report W grade or content, no reserves are reported, and recovery factors are not applied; they therefore do not establish a tungsten resource. [1,2,7,18-20]

Decision snapshot

Question	Assessment	Meaning for Μετάλλιον
Best nodule-first target	Cook Islands EEZ - Penrhyn Basin / high-abundance corridor	Best combination of mapped abundance, Co-rich chemistry, known W and current project MREs; no W resource grade is reported.
Highest nodule W assay	Indian Ocean sample CP8130: 148.5 ppm W at 13°35'S, 77°03'E; 5,270 m	High-value verification target, but a single sample cannot establish a district-scale W anomaly.
Highest marine Fe-Mn W result	Emperor Chain Guyot crusts: up to 287 ppm W	Higher assay, but a different deposit type and no demonstrated recoverable resource.
Best Ni-Cu nodule province	Clarion-Clipperton Zone (CCZ)	Strongest established nodule field for Ni-Cu-Co; not the best W-led target.
W-only investment case	Weak on current grade evidence	Marine samples are materially lower grade than typical terrestrial tungsten ores.

Three findings that should shape the strategy

- Geology: tungsten is mainly scavenged as tungstate onto Fe-Mn oxyhydroxides; sediment diagenesis can raise W in some nodule populations. It is not normally present as discrete wolframite or scheelite grains. [1,4]
- Grade: 59 ppm W in Cook Islands nodules equates to only about 0.0074% WO_3 equivalent. Typical skarn-hosted tungsten ores are commonly about 0.3-1.4% WO_3. A marine project must therefore be justified by the aggregate polymetallic basket and a proven recovery flowsheet. [9,18]
- Timing: the Cook Islands currently authorise exploration only; the ISA exploitation code for the international seabed remains under negotiation as of November 2026. [11-13]

Scope note: “tungsten polymetallic nodules” is used here as commercial shorthand. The documented geological material is ferromanganese (Fe-Mn) nodule material containing tungsten as a trace constituent, alongside more concentrated metals.

1. Geological model: where the tungsten sits

Marine polymetallic nodules are generally rounded Fe-Mn oxide concretions that grow at or just below the sediment-water interface on abyssal plains. Their main mineral hosts are manganese oxides, including vernadite/delta-MnO2 and related phases, together with poorly crystalline iron oxyhydroxides. Metals are scavenged from seawater and porewater and incorporated by adsorption, surface complexation, oxidation and structural incorporation. [3,4]

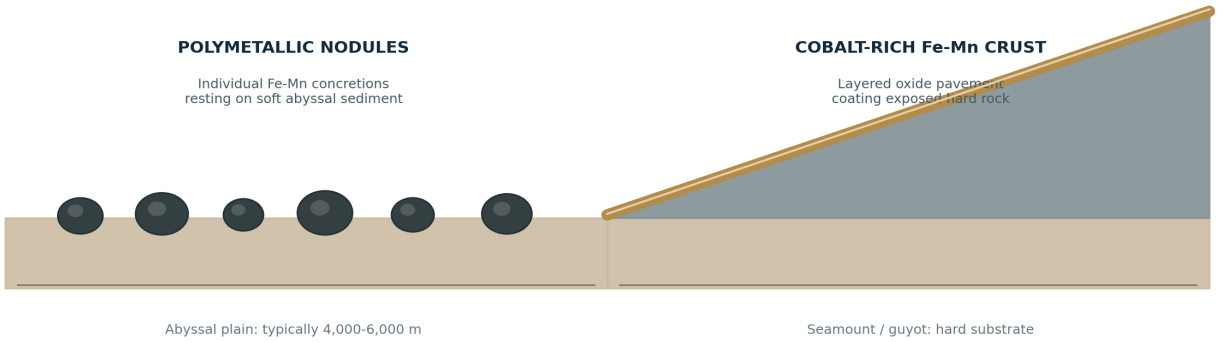


Figure 1. Schematic distinction between abyssal nodules and seamount ferromanganese crusts. The distinction matters: the highest W result found in this review is from a crust, not a nodule.

Nodule formation controls

Control	Favourable setting	Exploration implication
Sedimentation rate	Very low net sediment accumulation; nodules remain exposed at the seabed.	Prioritise flat, sediment-covered abyssal plains, not rapidly accumulating continental-margin muds.
Water depth	Most economically studied fields occur at roughly 4,000-6,000 m.	Requires deep-rated ROV/AUV, box coring and eventual high-pressure lift systems.
Nucleus supply	Small rock fragments, fossils, teeth or earlier nodules provide a nucleus.	Local substrate and sediment texture affect nodule size and distribution.
Redox / porewater	Hydrogenetic growth draws metals from seawater; diagenetic growth draws metals remobilised from sediment porewater.	The balance affects Co-Ni-Cu-Mn chemistry and can influence W enrichment.
Stable seafloor	Low disturbance and long periods of exposure favour growth.	Tectonically rough, volcanic or high-current environments may be poor for nodule blanket continuity.

How tungsten is captured

In oxic seawater, dissolved tungsten is predominantly present as tungstate (WO4 2-). Experimental and review literature indicates that W can be progressively enriched on Fe-Mn (oxy)hydroxides, particularly through surface-complexation processes associated with the iron-rich phases. Sediments can act as a reservoir: porewater release and early oxic diagenesis may increase the W available to Mn oxides. Rose-Koga and Albarede found a negative relationship between W and magnesium isotope composition in Indian Ocean nodules, interpreting the pattern as evidence for stronger early diagenesis in the W-rich samples. [1,4]

Geochemical interpretation

A W anomaly in a nodule is best treated initially as a process indicator - evidence of Fe-Mn sorption and/or diagenetic enrichment - rather than proof of a tungsten ore body. The key question is whether the W enrichment is laterally persistent, recoverable, and accompanied by credits from the other metals.

2. Global distribution and measured tungsten

Fe-Mn nodules occur in all major ocean basins, but the best-known large fields are the Clarion-Clipperton Zone (CCZ), Peru Basin, Penrhyn Basin / Cook Islands, and Central Indian Ocean Basin. Other regions include the western Pacific, the Atlantic, Southern Ocean and abyssal areas around island arcs and ridges. Global occurrence does not mean equivalent grade or resource confidence. [3,5]

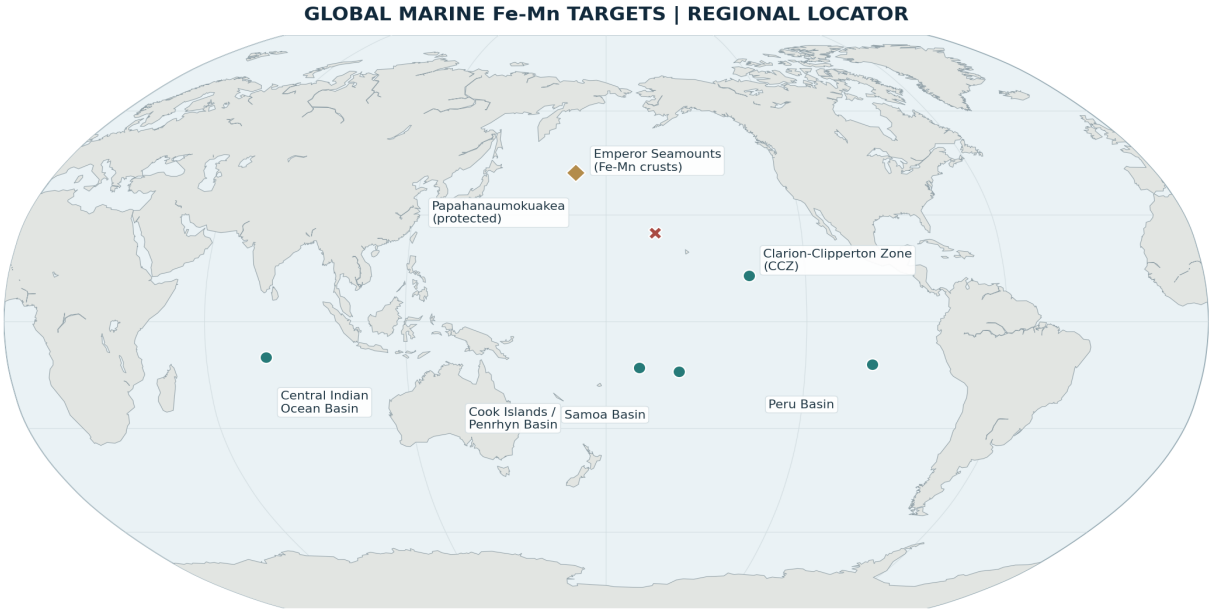


Figure 2. Global locator of the principal nodule and crust provinces discussed in this report. Broad field locations are schematic and are not legal boundaries or surveyed mineral footprints.

Published W concentrations

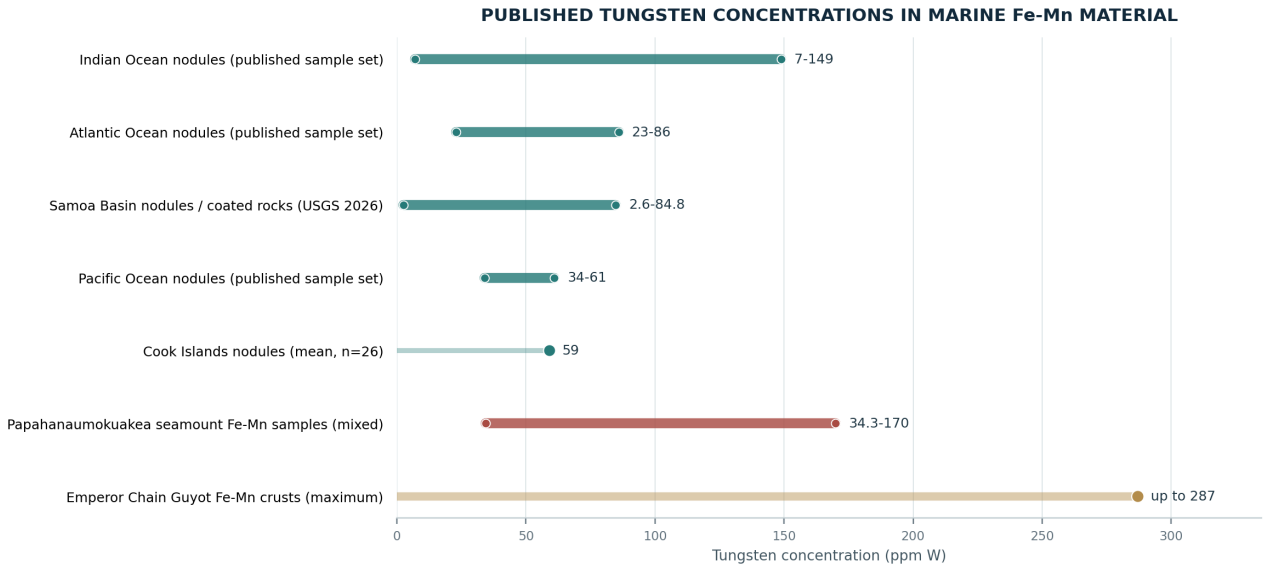


Figure 3. Tungsten content of published marine Fe-Mn samples. The 2010 cross-basin nodule ranges come from Rose-Koga and Albarede; Cook Islands mean is a separate 26-sample dataset; Samoa includes nodules and coated rocks; Emperor and Papahanaumokuakea are crust / seamount comparisons. [1,6-8,18]

The 2010 comparative dataset reports Pacific nodules at 34-61 ppm W, Indian Ocean nodules at 7-149 ppm, and Atlantic nodules at 23-86 ppm. The most W-rich sample in that table is CP8130 (148.5 ppm) from 13°35'S, 77°03'E at 5,270 m. A second high-W sample, DR8605, contains 139.2 ppm W at 58°02'S, 77°01'E and 2,600 m. These are sample observations, not continuous grade estimates. [1]

3. Assay evidence and priority sample points

The most useful publicly accessible W dataset for direct nodule comparison remains the cross-basin study by Rose-Koga and Albarede (2010). It contains sample coordinates, water depths and W assays, but the sample population is small and not designed as a modern resource survey. The Indian Ocean result is therefore a geochemical lead, not a mineral resource estimate. [1]

Sample / region	Latitude	Longitude	Depth (m)	W (ppm)	Selected associated chemistry / note
CP8130 - Indian Ocean	13°35' S	77°03' E	5,270	148.5	Mn 26.40%; Ni 1.258%; Cu 1.248%; sample is an unusually interesting W-Cu-Ni nodule.
DR8605 - Indian Ocean / Southern Ocean sector	58°02' S	77°01' E	2,600	139.2	Mn 21.10%; Co 1.055%; W-rich but the site is shallower and requires geological context checks.
AET7611 - Indian Ocean	24°05' S	59°55' E	4,425	89.5	Intermediate W anomaly; useful for checking regional continuity.
CP8123 - Indian Ocean	11°51' S	78°39' E	5,060	69.3	Broader Indian Ocean comparison; about 260 km from CP8130, not a local continuity control.
AET7610 - Indian Ocean	31°21' S	61°55' E	4,245	53.2	Moderate W assay; useful comparison point.
Cook Islands sample mean	-	-	-	~59	26-sample composition dataset; hydrogenetic nodule population. [18]
Samoa Basin (USGS 2026)	-	-	-	2.6-84.8	Range across sampled nodules and Fe-Mn coated rocks; modern geochemical dataset.

Interpretation of the Indian Ocean anomaly

CP8130 is the most compelling nodule sample for W-led follow-up because it combines 148.5 ppm W with high measured Ni and Cu in the published transition-element table. However, one sample can be influenced by a local nucleus, microfacies, diagenetic patch or sample preparation. The two CP samples are about 260 km apart and provide regional comparison points for a GIS re-check, not a local sample cluster; no claim of continuity should be made until new cores are collected. [1]

Cook Islands W inventory: a caution on historical estimates

Hein et al. (2015) used JICA / MMAJ survey data and geochemical analyses to map six potential high-abundance areas of roughly 20,000 km2 each. The selected northern area of the principal north-south abundance belt was modelled to contain 30,215 tonnes of W, alongside large modelled inventories of Mn, Ni, Mo, V and Zr. That W figure is a historical, extrapolated contained-metal estimate—not a current W-specific Mineral Resource or Reserve. Importantly, 2026 S-K 1300 reports for EL1 and EL3 disclose modern nodule resources with Co, Cu, Fe, Mn and Ni grades, but do not separately report W grade or contained W. The recent project MREs therefore support the broader nodule proposition, not a W resource claim. [2,19,20]

Critical due-diligence action

Acquire the raw sample-level Cook Islands dataset, the EL1 / EL3 block models and assay tables, and the Rose-Koga / Albarede table. Re-assay or request W data for the current resource domains, normalise assays to a dry basis, and compare W with nodule abundance. Separately assay Fe-Mn coating versus nodule interior, nucleus and sediment to isolate the true W host.

4. Global target ranking

Targets are ranked for a company looking for tungsten-bearing polymetallic nodules, with an emphasis on (i) observed W, (ii) nodule abundance, (iii) co-metal potential, (iv) data quality and (v) practical access. Rankings are strategic screening judgements, not resource classifications.

Rank	Target	W evidence	Other metals / geological case	Screening view
1	Cook Islands EEZ - Penrhyn / northern abundance corridor	Mean ~59 ppm W in a 26-sample dataset [18]; 2015 modelled 30,215 t contained W in one selected area [2]. 2026 EL1/EL3 MRE tables omit W [19,20].	High Co, Ti and REY; nodule abundance 19-45 kg/m2 across mapped belt, local patches to 58 kg/m2.	Best first nodule programme. W is a by-product thesis; current regime is exploration-only.
2	Central Indian Ocean Basin - CP8130 vicinity	Highest reported nodule assay in the cited dataset: 148.5 ppm W; a second sample, CP8123, 69.3 ppm roughly 260 km away.	Some samples contain material Ni-Cu and Mn grades; basin is a known mixed nodule province.	Best W-assay verification target. Spatial continuity and abundance need re-mapping.
3	Clarion-Clipperton Zone (CCZ)	Pacific nodules in the global dataset commonly 34-61 ppm W; no evidence here of a W-specific grade advantage.	Largest and most intensively explored Ni-Cu-Co nodule province; high nodule abundance and 17 ISA exploration contracts.	Strongest established polymetallic nodule ecosystem; pursue only as a multi-metal play.
4	Samoa Basin / American Samoa abyssal plains	USGS 2026 measurements: 2.6-84.8 ppm W in nodules and coated rocks.	Fresh 2026 mapping, abundance and geochemistry dataset; modern sample positions and abundance are available.	High value for science-led regional reconnaissance; early-stage in resource terms.
5	Peru Basin	No comparable W high-grade signal established in sources reviewed.	Classic diagenetic nodule field; generally prospective for Mn-Ni-Cu.	Useful comparator and possible mixed-metal field, not W-led.
6	Atlantic nodule provinces	Published nodule range 23-86 ppm W in limited samples.	Known occurrences off North America, Brazil, Argentina and western Africa; relatively underexplored.	Frontier search only; broad geochemical and abundance coverage is weak.

Separate marine crust targets - not nodule targets

Emperor Seamount Chain: a recent study reports W up to 287 ppm in Fe-Mn crusts from Jingū, Ōjin and Nintoku Guyots, with mean Mo of 879 ppm and high REY. The elevated W is a useful indication that certain western Pacific Fe-Mn growth environments can enrich W, but crusts grow on hard-rock seamount surfaces and have different geometry, thickness, sampling and recovery constraints. [7]

Papahānaumokuākea: USGS data report 34.3-170 ppm W in crusts, nodules and coated cobbles sampled from seamounts. This is an important geochemical comparator, but the Marine National Monument prohibits mining, drilling and exploration for oil or gas; it is not an actionable exploration recommendation. [8,10]

5. Priority area: Cook Islands EEZ

The Cook Islands offers the best current combination of nodule-scale evidence, abundance mapping, cobalt-rich chemistry and a national regulatory pathway. The EEZ includes the Penrhyn and Samoa abyssal basins, with large areas of exposed or partially buried manganese nodules. The 2015 USGS-indexed study describes a high-abundance belt running from the south-east of the EEZ toward the north-west, with a south-west branch. [2]

Legacy geochemical character: predominantly hydrogenetic nodules containing mean Co around 0.41%, Ni around 0.38%, Ti around 1.20% and total REY+Y around 0.167% in the 2015 study. The mean growth rate was estimated at 1.9 mm per million years. W, Mo, Nb, V and Zr were also identified as enriched trace metals. [2]

2026 field update

In its first 2026 Cook Islands ROV exploration leg, NOAA and Cook Islands scientists mapped 24,253 km2 of the EEZ and collected nodule, sediment, biological and water samples. Nodules were observed at seven dive locations, including an abyssal plain along the western EEZ boundary at about 5,600 m and an abyssal plain near the boundary with American Samoa at about 5,700 m. The western waters were previously poorly studied. NOAA explicitly notes that further analysis is required before resource potential can be established. [16]

Screening zone	Geological rationale	Evidence / next step
Penrhyn Basin	Established nodule-bearing abyssal basin; part of the hydrogenetic Cook Islands province.	Reconcile legacy survey grids with modern bathymetry and chemistry. [2]
Northern N-S abundance belt	Selected northern sector of the mapped high-abundance corridor; historical model optimised for multiple metals.	Revisit the 30,215 t contained-W estimate as a hypothesis, not a resource. [2]
Western EEZ abyssal plain	New nodule observations from the 2026 ROV work at about 5,600 m.	Acquire dive tracks, video annotations, sample coordinates and assays. [16]
American Samoa boundary basin	Nodules observed at about 5,700 m in 2026; likely valuable regional control.	Survey abundance and W grades independently; do not infer grade from visual cover. [16]

Table 2. Priority Cook Islands screening sectors. This is a qualitative index, not a map, licence boundary or surveyed mineral footprint.

Contemporary project-level resources (S-K 1300; effective 31 Dec 2025)

EL1 / CIC: 1,950 Mt wet Inferred at a 13 kg/m² abundance cut-off. EL3 / Moana: 417 Mt wet Indicated plus 102 Mt wet Inferred at a 12.5 kg/m² cut-off. These are separate licence-area estimates, not additive regional tonnage. The published resource tables report Co, Cu, Fe, Mn and Ni, not W; neither report establishes a W resource. No Mineral Reserves or completed project cash-flow analysis are disclosed. [19,20]

Recommended local targeting logic

- First pass: prioritise the Penrhyn Basin and the northern section of the N-S high-abundance belt, where the 2015 study placed its multi-metal-optimised W inventory. [2]
- Second pass: incorporate the 2026 western EEZ mapping and ROV dives, particularly the western boundary abyssal plain and the 5,700 m basin close to American Samoa. Do not assume W grade from nodule presence alone. [16]
- Third pass: distinguish nodule-covered abyssal plain from ferromanganese-coated basalt/seamount material; both were observed in the western Cook Islands. [16]

Best current company thesis

A Cook Islands project should be underwritten as a cobalt-rich polymetallic nodule opportunity with W, Mo, V, Zr and REY as potential by-products. The 30,215 t modelled W inventory is an exploration hypothesis to verify, not an investment-grade resource statement.

6. Secondary targets and global comparators

Central Indian Ocean Basin

The strongest nodule W assay in the cited cross-basin dataset is CP8130 at 148.5 ppm W, 13°35'S, 77°03'E, depth 5,270 m. The same sample contains 1.258% Ni and 1.248% Cu in the transition-element dataset, making the occurrence more interesting as a Cu-Ni-W-bearing Fe-Mn sample than as a W-only target. Another CP-series sample, CP8123 at 11°51'S, 78°39'E and roughly 260 km away, returned 69.3 ppm W. This is enough to justify a modest verification campaign, but not enough to infer a continuous high-grade zone. [1,5]

Recommended work: recover georeferenced expedition/sample metadata; confirm that sample locations align with current bathymetry and the modern nodule-field boundary; collect duplicate box-core / multi-core samples across a grid; assay nodules, sediments and nuclei separately; test the lateral persistence of W and the relation to Fe/Mn ratio and diagenetic indicators.

Clarion-Clipperton Zone

The CCZ is the world’s most advanced nodule exploration province and one of the strongest global fields for Ni-Cu-Co. ISA describes nodules as most abundant in the CCZ, the Central Indian Ocean Basin and Peru Basin. As of 29 October 2026, the ISA had 31 active exploration contracts in areas beyond national jurisdiction, including 17 in the CCZ. The W data cited here do not show a distinctive CCZ W grade advantage; its relevance is the commercial and geological maturity of the broader nodule system. [5,11]

Peru Basin and South Pacific

The Peru Basin is a classic diagenetic nodule end-member, while the CCZ and Central Indian Ocean Basin are commonly treated as mixed hydrogenetic-diagenetic. Genetic type influences the wider metal suite and nodule texture; in some Indian Ocean samples, stronger early diagenesis is associated with higher W. The Peru Basin remains an important Ni-Cu-Mn comparison province, but this review did not identify a compelling W-specific assay or resource case there. [1,5]

Samoa Basin: new data, early stage

A 2026 USGS field campaign collected nodules and Fe-Mn-coated rocks in the Samoa Basin and released sample locations, depths, abundance and geochemical data. Its W range is 2.6-84.8 ppm. The dataset is strategically useful because it is recent and pairs chemistry with abundance measurements, but it does not establish a high-W resource. It may provide a modern regional baseline and identify where W, Co, Ni, Cu and REY co-vary with nodule density. [6]

Ranked fieldwork logic

Cook Islands first for integrated nodule prospectivity; Central Indian Ocean second for validating the strongest W assays; Samoa Basin as a data-rich regional reconnaissance play; CCZ as a reference province or partnership market for multi-metal nodules; Peru Basin as a genetic and metallurgical comparator.

7. The high-W marine exception: cobalt-rich crusts

Ferromanganese crusts grow as coatings on exposed hard rock, particularly on seamounts and guyots. They form by slow hydrogenetic precipitation from seawater and can concentrate Co, Te, REY, Mo, W and other trace metals. They are not equivalent to nodules: their geometry is a thin coating on steep or irregular rock, often at shallower depths than abyssal nodule fields. [3,5]

Emperor Chain Guyots - Jingū, Ōjin and Nintoku

A recent study of Fe-Mn crusts from the central Emperor Chain reports tungsten up to 287 ppm, roughly double the levels cited for other global ocean regions in that paper. The same crust population shows high molybdenum (average 879 ppm; maximum 1,429 ppm), lead (average 3,029 ppm; maximum 5,303 ppm) and total REY up to 5,431 ppm. The authors interpret the crusts as primarily hydrogenetic with a hydrothermal contribution. This is the highest W result identified in this review for marine Fe-Mn material, but it is a crust assay and does not demonstrate thickness, areal continuity, tonnage or recoverability. [7]

Papahānaumokuākea - useful comparator, prohibited target

USGS data for seamount crusts, nodules and coated cobbles from the Papahānaumokuākea Marine National Monument show W from 34.3 to 170 ppm. The concentration range provides a second western Pacific benchmark, but the monument explicitly prohibits mining and exploration for oil or gas. No operational target recommendation is made for this protected area. [8,10]

Implications for Μετάλλιον Inc.

- Keep the Emperor Chain on a separate “crust” track. A W concentration above 200 ppm may be worth detailed geochemical work, but only if crust thickness, density, Co-Te-REY content, substrate, slope and ecosystem constraints can be established.
- Do not use crust assays to set nodule grade expectations. Crusts and nodules have different growth rates, host phases, physical geometry and collection methods.
- Avoid protected areas. Use the Papahānaumokuākea dataset only as a scientific benchmark for regional geochemical variability.

Interpretation

The Emperor Chain is the leading marine Fe-Mn W assay anomaly in this review, but it is a research target rather than a current commercial tungsten resource. For a nodule-focused programme, the Cook Islands and Indian Ocean remain the relevant choices.

8. Metallurgy, grade and economic reality

W concentration conversion

The conversion factor from elemental tungsten to tungsten trioxide is approximately 1.261. The table below converts the observed marine W values into WO3-equivalent concentration for comparison with terrestrial tungsten ores. It is a chemical conversion only; it does not imply that the W is recoverable as a saleable concentrate.

Material / assay	W content	WO3 equivalent	WO3 per dry tonne
Cook Islands nodule mean	59 ppm W	0.0074 wt% WO3	~0.074 kg/t
Indian Ocean nodule high sample	149 ppm W	0.0188 wt% WO3	~0.188 kg/t
Emperor Chain crust maximum	287 ppm W	0.0362 wt% WO3	~0.362 kg/t
Typical tungsten skarn ore range	0.3-1.4 wt% WO3	0.3-1.4 wt% WO3	3-14 kg/t

The concentration gap is material

At the Cook Islands mean, each dry tonne of nodules contains only about 59 grams of elemental W, equivalent to roughly 74 grams of WO3. Even the 149 ppm nodule sample equates to around 0.019% WO3. By comparison, published typical grades for tungsten skarn deposits are about 0.3-1.4% WO3, while porphyry examples are commonly 0.1-0.4%. These differences are substantial: approximately 40-190 times lower for the Cook Islands mean versus skarn grade, and about 16-75 times lower for the 149 ppm sample. [9]

Supply context: the USGS estimates world mine production of about 85,000 tonnes of contained tungsten in 2025, including about 67,000 tonnes from China. This concentration is strategic context, not evidence that low-grade nodules can meet tungsten demand; the historical 30,215-tonne Cook Islands model is neither annual supply nor a recoverable inventory. [2,17]

Recovery route: research hypothesis, not proven flowsheet

Because tungsten is likely held by sorption or structural incorporation in Fe-Mn oxyhydroxides, a commercial flowsheet would need to demonstrate that W can be selectively released without consuming excessive reagents or compromising recovery of the principal metal suite. A sensible test programme would compare reductive and oxidative leaching, pH-controlled desorption, Fe/Mn phase dissolution, impurity deportment and W recovery to a purified tungstate product. There is no basis in the sources reviewed to assume a mature, commercially demonstrated nodule-to-W process.

Recommended metallurgical test matrix

Test	Question answered	Decision use
Mineralogical deportment	Is W associated with FeOOH, Mn oxide, sediment, nucleus or discrete grains?	Choose leach chemistry; avoid relying on bulk assay alone.
Sequential leach / desorption	What fraction is readily leachable and at what pH / redox?	Establish extraction ceiling and reagent intensity.
Co-metal extraction	Can W be captured from the process liquor after Ni, Co, Cu, Mn and Fe handling?	Determine if W is a low-cost by-product or a process penalty.
Impurity control	How much P, As, Si, Fe, Mn and other impurities report to W liquor?	Estimate purification burden and product quality.
Residue / water balance	What are solid residue chemistry, neutralisation and recycle loads?	Test environmental and operating cost assumptions.

Commercial conclusion

A tungsten-only seabed mining case is not supported by current grade evidence. The rational strategy is a multi-metal nodule system in which tungsten may be recovered as a minor by-product if and only if the primary metal flowsheet already works and W recovery adds net value.

9. Proposed exploration programme

A staged, low-regret programme can determine whether the W signal is reproducible before Μετάλλιον Inc. commits to major survey or vessel expenditure. Work should be designed to separate three uncertainties: grade, abundance and recoverability.

Stage	Work package	Deliverables	Gate to proceed
0 - Data room	Acquire raw published geochemistry, sample metadata, bathymetry, nodule abundance grids, dredge/core logs, licence and protected-area boundaries.	GIS prospectivity model; source register; QA/QC audit; data gaps.	Coordinates and sample IDs reconcile; no critical data provenance gaps.
1 - Assay verification	Reassay archived material where accessible; repeat ICP-MS / XRF for W, Mo, V, Co, Ni, Cu, Mn, Fe, REY, Te, Nb, Zr; use standards and duplicates.	Validated assay dataset on dry basis; mineralogical deportment by XRD / SEM-EDS / LA-ICP-MS where useful.	W anomaly survives duplicate and laboratory verification.
2 - Targeted seafloor sampling	Box cores / multi-cores in a grid around CP8130 and Cook Islands priority corridors; ROV visual transects; backscatter and high-resolution bathymetry.	Abundance (kg/m2), nodule size distribution, sedimentology, nodule chemistry, habitat baseline.	W is spatially repeatable and co-located with nodule abundance.
3 - Metallurgical scoping	Bench-scale leach and separation tests on representative composite samples; full mass balance and residue characterisation.	Extraction and impurity profiles; reagent consumption; preliminary product spec and residue plan.	Net co-product value is technically plausible on a multi-metal basis.
4 - Pre-feasibility screen	Compare vessel, lift system, wet-tonnage, transport, processing, energy, environmental monitoring and closure assumptions.	Conceptual project model; risk register; independent technical review.	Only proceed if environmental, regulatory and economic hurdles can all be cleared.

Priority work areas

- Cook Islands: sample the north-south abundance belt, the northern high-abundance / W-inventory area described in the 2015 study, and western abyssal plains highlighted by the 2026 NOAA expedition. Reconcile EL1 / EL3 block models and obtain W assays, which are absent from the published resource tables. [19,20]
- Indian Ocean: re-locate CP8130 and CP8123 using the original cruise metadata; design a fine-scale sampling grid around the 148.5 ppm and 69.3 ppm points and a broad regional control transect.
- Samoa Basin: ingest the 2026 USGS data release because it pairs chemistry, location, depth and nodule abundance; use it as a regional exploration dataset and geochemical control.
- Emperor Chain: if considered, develop a separate crust prospect model with thickness and substrate mapping; do not mix it into nodule tonnage assumptions.

Suggested analytics stack

Build a consistent GIS database using WGS84 coordinates and a dry-weight analytical basis. Link nodule abundance (kg/m2), seafloor slope, depth, sedimentation rate, bottom-water oxygen, carbonate compensation depth, Fe/Mn ratio, Mn mineralogy and porewater proxies to W assays. Apply a spatially blocked validation design: random train/test splits can overstate continuity where samples are spatially clustered. All interpolated W surfaces should show prediction uncertainty and be labelled as exploration models.

10. Regulatory, environmental and execution constraints

Cook Islands EEZ

The Cook Islands Seabed Minerals Authority states that the country currently allows exploration only and has three exploration licences for polymetallic nodules; commercial harvesting has not been authorised or commenced. The 2026 Cook Islands-US framework supports potential cooperation on research, mapping, investment discussion, processing and supply-chain security, but the SBMA explicitly states that the framework is not a mining licence or approval and does not commit the country to extraction. Exploration title does not imply permission to harvest nodules. [12,13]

Areas beyond national jurisdiction

For UNCLOS parties, mineral activity in the Area beyond national jurisdiction is regulated by the International Seabed Authority. In September 2026 the ISA reported that exploitation-regulation negotiations had advanced, but the Council was still seeking a roadmap and revised negotiating text for the 32nd Session; negotiations were scheduled to resume in March 2027. The October 2026 Congressional Research Service update reported 31 active ISA exploration contracts, 17 in the CCZ, and no exploitation contracts. [11,14]

The United States has a separate domestic framework under the Deep Seabed Hard Mineral Resources Act for U.S. citizens and companies in areas beyond national jurisdiction. The 2026 NOAA rule clarifies the exploration and commercial-recovery permit process. This is a distinct legal pathway and does not itself resolve international recognition or conflicts with the UNCLOS regime. [15]

Environmental constraints

- Nodule growth is exceptionally slow: the Cook Islands study reports a mean of 1.9 mm per million years. Removed nodules will not be replenished on human timescales. [2]
- Nodules are hard substrate for specialised abyssal organisms. Collection can remove habitat, disturb sediment and create sediment plumes; baseline biological and plume studies are essential before any recovery proposal. [3,9]
- The 2026 NOAA Cook Islands cruise explicitly collected biological samples, water samples and environmental DNA alongside nodules, demonstrating the baseline-data model required for responsible decision-making. [16]
- Papahānaumokuākea Marine National Monument prohibits mining and exploration for oil or gas. The high W assay benchmark from this area is not a lawful prospecting recommendation. [10]

Key execution risks

Risk	Why it matters	Mitigation
Grade risk	Sparse samples and patchy diagenetic enrichment can overstate true regional grade.	Dense, spatially blocked sampling; separate nodule / crust / sediment assays.
Resource risk	Historical EEZ-wide and W-contained estimates are not substitutes for project MREs; even current MREs are not reserves and do not prove recoverability.	Independent technical review; apply CRIRSCO/JORC/CIM principles only after adequate data.
Metallurgy risk	W may be strongly bound or diluted through Fe-Mn processing.	Early deportment and leach tests on representative composites.
Legal risk	Exploration title, commercial recovery permit, environmental consent and international recognition are distinct.	Local counsel; regulator engagement; avoid relying on political statements as permits.
ESG / social licence	Deep-sea biodiversity and long recovery times pose high ecological and reputational risk.	Open baseline science, precautionary thresholds, independent monitoring and stakeholder consultation.

11. Final findings and recommendation

Finding 1 - Tungsten is real, but not a primary marine nodule metal

The marine Fe-Mn literature documents W in nodules across the Pacific, Indian and Atlantic oceans. The measured levels in the cited nodule datasets are commonly tens of ppm and can reach about 149 ppm in individual Indian Ocean samples. W is carried largely by Fe-Mn oxyhydroxide scavenging and diagenetic processes, rather than forming a discrete tungsten ore mineral assemblage. [1,4]

Finding 2 - Cook Islands is the best nodule-first location

The Penrhyn Basin / Cook Islands EEZ combines extensive mapped nodule abundance, high cobalt and critical-metal content, published W-bearing nodule data and a current national exploration framework. The 2026 NOAA ROV expedition adds new mapping and confirmed nodule observations in areas previously poorly sampled. The 2015 contained-W estimate of 30,215 tonnes in a selected area is a useful hypothesis to verify, not a W resource statement. Recent EL1 and EL3 S-K 1300 reports establish project nodule resources, but do not report W in their resource tables. [2,12,16,18-20]

Finding 3 - Indian Ocean offers the strongest nodule assay lead

The CP8130 sample at 13°35'S, 77°03'E is the most compelling published nodule W assay located in this review. Its associated Ni-Cu values merit re-analysis. The sample remains a point observation; continuity and abundance must be demonstrated by new sampling. [1]

Finding 4 - Emperor Chain is a separate high-W crust opportunity

Emperor Chain Fe-Mn crusts yield the highest published W concentration found in this review, up to 287 ppm. This should remain a separate crust research track, not be labelled a nodule deposit or counted toward nodule tonnage. [7]

Finding 5 - W-only economics are not supported

The elemental W grade in nodules is several orders below a tungsten concentrate and is much lower than typical primary tungsten ore grades. Any value proposition requires a technically proven multi-metal process and a credible environmental and legal pathway. [9]

> **Board-level decision**
> Approve a desktop data and assay-verification phase focused on (1) Cook Islands Penrhyn / abundance corridor and (2) Indian Ocean CP8130 follow-up. Do not approve a W-only resource claim, a deep-sea mining capex case or a commercial valuation until the W deportment, recovery, nodule abundance, biodiversity baseline and applicable approvals are independently demonstrated.

What would change the conclusion?

A repeatable W anomaly above 150-200 ppm across multiple spatially distributed nodule samples, high areal abundance, demonstrated leach recovery, and positive net value within an already viable Co-Ni-Cu-Mn flowsheet would materially improve the case. This is a proposed diligence trigger, not an established economic cut-off. Conversely, if W is isolated to individual nodules or is not selectively recoverable, it should be omitted from project economics.

References and source register

Primary geological and geochemical sources

[1] Rose-Koga, E.F. & Albarede, F. (2010). A data brief on magnesium isotope compositions of marine calcareous sediments and ferromanganese nodules. Geochemistry, Geophysics, Geosystems, 11. DOI: 10.1029/2009GC002899. Includes sample coordinates, depths and W assays; reports Pacific 34-61 ppm, Indian 7-149 ppm and Atlantic 23-86 ppm. [source]

[2] Hein, J.R., Spinardi, F., Okamoto, N., Mizell, K., Thorburn, D. & Tawake, A. (2015). Critical metals in manganese nodules from the Cook Islands EEZ, abundances and distributions. Ore Geology Reviews. DOI: 10.1016/j.oregeorev.2014.12.011. USGS publication summary includes mapped abundance, chemistry and modelled W inventory. [source]

[3] Hein, J.R., Koschinsky, A. & Kuhn, T. (2020). Deep-ocean polymetallic nodules as a resource for critical materials. Nature Reviews Earth & Environment, 1, 158-169. DOI: 10.1038/s43017-020-0027-0. [source]

[4] Enrichment Characteristics and Mechanisms of Critical Metals in Marine Fe-Mn Crusts and Nodules: A Review (2023). Minerals, 13(12), 1532. Review of hydrogenetic, diagenetic and mixed enrichment mechanisms, including Mo-W-V adsorption and typical W values below 100 ppm. [source]

[5] Mizell, K., Hein, J.R., Au, M.V. & Gartman, A. (2022). Estimates of metals contained in abyssal manganese nodules and ferromanganese crusts in the global ocean based on regional variations and genetic types of nodules. DOI: 10.1007/978-3-030-87982-2_3. [source]

[6] Adamczyk, K.B. et al. (2026). Abundance and geochemistry of abyssal ferromanganese nodules and ferromanganese coated rocks from the Samoa Basin (2026-604-FA). USGS data release, 23 June 2026. DOI: 10.5066/P149BW9Z. [source]

[7] Compositional variation and genesis of ferromanganese crusts in the central segment of Emperor Chain Guyots, Pacific Ocean (2025/26 publication record). Reports W up to 287 ppm in Fe-Mn crusts from Jingū, Ōjin and Nintoku Guyots. [source]

[8] Favela, J.J. et al. (2024). Geochemistry of ferromanganese crusts, nodules, and coated cobbles from seamounts in the Papahānaumokuākea Marine National Monument. USGS data release, DOI: 10.5066/P1WYF9T7. [source]

References and source register (continued)

Economic, exploration, regulatory and environmental sources

[9] Frontiers in Chemistry (2020). The Challenge of Tungsten Skarn Processing by Froth Flotation: A Review. Typical tungsten skarn grades reported at 0.3-1.4% WO3. DOI: 10.3389/fchem.2020.00230. [source]

[10] Papahānaumokuākea Marine National Monument. Protection measures: all extractive activities restricted; no mining, drilling or exploration for oil or gas. [source]

[11] U.S. Congressional Research Service (29 October 2026). U.S. Interest in Seabed Mining in Areas Beyond National Jurisdiction: Brief Background and Recent Developments. Reports 31 active ISA exploration contracts, 17 in the CCZ, and no exploitation contracts. [source]

[12] Cook Islands Seabed Minerals Authority. Seabed Minerals Licensing Register. States that the Cook Islands allows exploration only and currently has three exploration licences for polymetallic nodules. [source]

[13] Cook Islands Seabed Minerals Authority. Cook Islands-US Framework for Securing of Supply in the Mining and Processing of Critical Minerals and Rare Earths (2026). Non-binding framework; not a mining licence or approval. [source]

[14] International Seabed Authority (19 September 2026). From the 31st to the 32nd Session: advancing the Mining Code and strengthening the Common Heritage of Humankind. Negotiations to resume in March 2027. [source]

[15] NOAA / Federal Register (21 January 2026). Deep Seabed Mining: Revisions to Regulations for Exploration License and Commercial Recovery Permit Applications, 91 FR 2642. [source]

[16] NOAA Ocean Exploration (7 August 2026). Marine Critical Minerals in the Cook Islands: A Preliminary Look. Reports 24,253 km2 mapped; nodules observed at seven dive locations in the 2026 ROV exploration leg. [source]

[17] U.S. Geological Survey (2026). Mineral Commodity Summaries 2026 - Tungsten chapter. Reports estimated world mine production of 85,000 t W in 2025, including 67,000 t from China; world reserves >4.7 Mt. [source]

[18] Pacific Community (SPC), STAR Abstracts (2013), PR175. Cook Islands nodule geochemistry abstract; reports mean W of about 59 ppm across 26 samples. Treat as a small legacy sample dataset, not a resource-grade statistic. [source]

[19] CIC Limited / RSC Consulting (2026). Initial Assessment and Technical Report Summary for the EL1 Cook Islands Polymetallic Nodule Deposit. S-K 1300 Mineral Resource effective 31 December 2025; revised 3 September 2026. Reports 1,950 Mt wet Inferred at 13 kg/m2 cut-off; resource table does not report W. [source]

[20] Moana Minerals Ltd / RSC Consulting (2026). Mineral Resource Estimate and Technical Report Summary for the EL3 Cook Islands Polymetallic Nodule Deposit. S-K 1300 Mineral Resource effective 31 December 2025; revised 3 September 2026. Reports 417 Mt wet Indicated plus 102 Mt wet Inferred at 12.5 kg/m2 cut-off; resource table does not report W. [source]

Methodology and limitations

This is a desktop geological screening report prepared from publicly available scientific papers, government data releases, regulator pages and technical reviews available as of 02 November 2026. W concentration ranges are not harmonised resource estimates: sample numbers, deposit types, analytical methods, moisture correction and sample populations differ. No independent fieldwork, assay verification, seafloor survey, metallurgical testwork, cost study or title due diligence has been performed for this report. Contained-metal estimates cited from historical studies are not Mineral Resources or Ore Reserves under JORC, NI 43-101, CIM or CRIRSCO reporting standards. No valuation, investment recommendation or assurance of legal access is provided.

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